SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving

Course Code:	DSA0405	**Course Title:**	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 –Industry Problem Solving	**Assessment:**	Assessment Tool 2– Industry Problem Solving
Weightage:	40% of CIA	**Total Marks:**	100
CO Statement:	Apply statistical estimation and hypothesis-testing techniques to solve realworld industry problems involving confidence intervals and inferential decisionmaking. (BL4)		
Student Name:	M V RAGHAVAN	**Reg. No.:**	192424373
Section / Batch:	AIDS	**Date:**	15-08-2026

Code of Conduct

I M V RAGHAVAN (192424373) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______________________________

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**	**Marks**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Problem 1 — Software Industry: API Response Time

A software company measures the response time of 100 API requests. The sample mean is 240 ms and standard deviation is 35 ms. Construct a 95% confidence interval for the average API response time.

1. Population and Sample

Population: the response times of all API requests handled by the company's service (past, present, and future). Sample: the 100 API requests measured in this study.

2. Parameter of Interest

The population mean response time, μ (in milliseconds).

3. Point Estimate $\bar{x} = 240$ *ms* *4–5. Hypotheses*

This is a pure estimation problem — no specific claimed or historical response-time target is given to test against. There is no meaningful H_0 / H_1 to state here; the goal is simply to estimate μ with a confidence interval. (If the company had an SLA target, e.g. "response time should not exceed 250 ms," the same interval could then be used to test H_0: μ = 250 vs H_1: μ ≠ 250.)

6. Significance Level α = 0.05 (for a 95% confidence interval).

7. Confidence Interval

$$CI = \bar{x} \pm z_{0.025} \times (s / \sqrt{n}) = 240 \pm 1.96 \times (35 / \sqrt{100}) = 240 \pm 6.86$$

95% Confidence Interval: (233.14 ms, 246.86 ms)

8–9. p-value and Comparison with α

Not applicable — with no hypothesized benchmark value, there is no test statistic or p-value to compute here (see step 4–5).

10. Industry-Oriented Conclusion

The company can be 95% confident that the true average API response time lies between 233.14 ms and 246.86 ms. This tight, wellestimated range (a margin of error of under 7 ms) gives the engineering team a reliable baseline for monitoring performance, setting realistic SLA targets, and detecting future degradation in response time.

Problem 2 — E-commerce: Conversion Rate

An e-commerce platform finds that 420 out of 600 randomly selected visitors purchased a product. Construct a 95% confidence interval for the true conversion rate.

1. Population and Sample

Population: all visitors to the e-commerce platform. Sample: the 600 randomly selected visitors observed.

2. Parameter of Interest

The population proportion of visitors who purchase a product, p (the true conversion rate).

3. Point Estimate $\hat{p} = x / n = 420 / 600 = 0.70$ *(70%)*

4–5. Hypotheses

As with Problem 1, no target conversion rate is specified to test against — this is a pure estimation task, so no H_0 / H_1 applies. (Were a target given, e.g. "industry average conversion is 65%," this same interval would support H_0: p = 0.65 vs H_1: p ≠ 0.65.)

6. Significance Level α = 0.05 (for a 95% confidence interval).

7. Confidence Interval

$$CI = \hat{p} \pm z_{0.025} \times \sqrt{[\hat{p}(1-\hat{p})/n]} = 0.70 \pm 1.96 \times 0.0187 = 0.70 \pm 0.0367$$

95% Confidence Interval: (66.33%, 73.67%)

8–9. p-value and Comparison with α

Not applicable — no hypothesized benchmark rate was provided (see step 4–5).

10. Industry-Oriented Conclusion

The platform can be 95% confident that its true conversion rate lies between 66.33% and 73.67%. This is a strong, tightly bounded conversion rate; marketing and product teams can use this range to set realistic KPI targets and to judge whether future campaigns produce a statistically meaningful lift beyond this baseline.

Problem 3 — Telecom: Monthly Data Consumption

A telecom company wants to estimate the average monthly data consumption of its users. A sample of 225 users has a mean consumption of 14.8 GB and standard deviation of 4.5 GB. Construct a 99% confidence interval for the population mean.

1. Population and Sample

Population: all users of the telecom company. Sample: the 225 users whose monthly data usage was measured.

2. Parameter of Interest

The population mean monthly data consumption, μ (in GB).

3. Point Estimate *$\bar{x}$ = 14.8 GB* ***4–5. Hypotheses***

No specific claimed average usage is given to test against, so this remains a pure estimation problem — no H_0 / H_1 applies. (A claim such as "average usage is 15 GB" could be tested against this same interval if provided.)

6. Significance Level α = 0.01 (for a
99% confidence interval).

7. Confidence Interval

$$CI = \bar{x} \pm z_{0.005} \times (s / \sqrt{n}) = 14.8 \pm 2.576 \times (4.5 / \sqrt{225}) = 14.8 \pm 0.773$$

99% Confidence Interval: (14.03 GB, 15.57 GB)

8–9. p-value and Comparison with α

Not applicable — no hypothesized usage benchmark was provided (see step 4–5).

10. Industry-Oriented Conclusion

The telecom company can be 99% confident that true average monthly data consumption across its user base lies between 14.03 GB and 15.57 GB. This estimate can directly inform data-plan design, network capacity planning, and pricing tiers, with a high level of statistical confidence appropriate for infrastructure-level decisions.

Problem 4 — Manufacturing: Component Diameter

A machine historically produces components with a mean diameter of 50 mm. A sample of 36 components has a mean diameter of 50.8 mm and standard deviation of 2.4 mm. Using a 95% confidence interval, determine whether 50 mm is a plausible population mean.

1. Population and Sample

Population: all components produced by the machine. Sample: the 36 components measured.

2. Parameter of Interest

The population mean component diameter, μ (in mm).

3. Point Estimate *$\bar{x}$ = 50.8 mm* ***4. Null Hypothesis (H_0)***

H_0: μ = 50 mm (the machine's true mean diameter still matches its historical value).

5. Alternative Hypothesis (H_1)

H_1: μ ≠ 50 mm (the machine's true mean diameter has shifted away from 50 mm).

6. Significance Level α = 0.05 (two-tailed test, for a
95% confidence interval).

7. Confidence Interval and Test Statistic

With n = 36 and population σ unknown (estimated by s), the t-distribution with df = 35 is used.

$$SE = s / \sqrt{n} = 2.4 / \sqrt{36} = 0.40$$

$$CI = \bar{x} \pm t_{0.025,35} \times SE = 50.8 \pm 2.030 \times 0.40 = 50.8 \pm 0.812$$

95% Confidence Interval: (49.99 mm, 51.61 mm)

$$t = (\bar{x} - \mu_0) / SE = (50.8 - 50) / 0.40 = 2.00$$

8. p-value

For t = 2.00 with df = 35 (two-tailed): p-value ≈ 0.0533.

9. Compare p-value with α p-value (0.0533) > α (0.05). Equivalently, the hypothesized value μ_0 = 50 mm falls just inside the 95% confidence interval (49.99, 51.61). We therefore fail to reject H_0.

10. Industry-Oriented Conclusion

There is not quite enough statistical evidence, at the 5% significance level, to conclude the machine's mean diameter has drifted from its historical 50 mm setting — though the result (p ≈ 0.053) is very close to the threshold. In a manufacturing quality-control context, this is a borderline result: the process may still be considered in control, but given how close 50 mm sits to the edge of the interval, it would be prudent to flag this machine for closer monitoring or a larger follow-up sample rather than dismissing the possibility of drift outright.

Problem 5 — Banking: Loan-Processing Time

A bank claims that its average loan-processing time is 5 days. A sample of 49 applications has an average processing time of 5.6 days with standard deviation 1.4 days. Construct a 95% confidence interval and determine whether the company's claim is reasonable.

1. Population and Sample

Population: all loan applications processed by the bank. Sample: the 49 applications whose processing time was recorded.

2. Parameter of Interest

The population mean loan-processing time, μ (in days).

3. Point Estimate $\bar{x} = 5.6$ *days* *4. Null Hypothesis (H_0)*

H_0: $\mu = 5$ days (the bank's claim is accurate).

5. Alternative Hypothesis (H_1)

H_1: $\mu \neq 5$ days (the true average processing time differs from the claimed 5 days).

6. Significance Level $\alpha = 0.05$ (two-tailed test, for a 95% confidence interval).

7. Confidence Interval and Test Statistic

With n = 49 and population σ unknown (estimated by s), the t-distribution with df = 48 is used.

$$SE = s / \sqrt{n} = 1.4 / \sqrt{49} = 0.20$$

$$CI = \bar{x} \pm t_{0.025,48} \times SE = 5.6 \pm 2.011 \times 0.20 = 5.6 \pm 0.402$$

95% Confidence Interval: (5.20 days, 6.00 days)

$$t = (\bar{x} - \mu_0) / SE = (5.6 - 5) / 0.20 = 3.00$$

8. p-value

For t = 3.00 with df = 48 (two-tailed): p-value ≈ 0.0043.

9. Compare p-value with α

p-value (0.0043) < α (0.05). Equivalently, the claimed value $\mu_0 = 5$ days falls outside the 95% confidence interval (5.20, 6.00). We therefore reject H_0.

10. Industry-Oriented Conclusion

There is strong statistical evidence that the bank's claimed 5-day average loan-processing time understates the true processing time. The data indicate the actual average is more likely between 5.2 and 6.0 days — meaningfully slower than advertised. From a business standpoint, the bank should either revise its public turnaround-time claim to avoid misleading customers and regulators, or investigate and address the operational bottlenecks causing the extra processing time.